

```
lo: flags=73<UP,LOOPBACK,RUNNING>mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 0 (Local Loopback)
    RX packets 19736 bytes 1686740 (1.6 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 19736 bytes 1686740 (1.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0
collisions 0
```

From the above listing, one can identify some of the basic network configuration elements, as shown below:

```
IP/netmask: 172.16.4.120/22
Mask: 255:255:252.0
Network Subnet ID: 172.16.4.0
First usable: 172.16.4.1
Last usable: 172.16.7.254
Broadcast: 172.16.7.255
Maximum Subnets: 64
Hosts per Subnet: 1022
Host Address Range: 172.16.4.1 - 172.16.7.254
```

The *netmask* parameter is particularly important, since it alone can tell you the size of your immediate network. In this case, 255.255.252.0 equates to four Class B addresses, because the difference of maximum number of hosts (256) and the number of masked hosts (252) is four. By combining the *netmask* with the configured IP address, it is possible to guess the range of the IP addresses in the local network. Because IP blocks are usually divided by whole groups and in sequence, you can tell that the IP address span of the network is from 172.16.4.1 to 172.16.7.254. The network and broadcast addresses are 172.16.4.0 and 172.16.7.255, respectively. A possible schematic diagram of the network is shown in Figure 1.

If there are multiple network devices, then *ifconfig* will return information for each device in the output, and from each interface it is possible to reach all the connecting networks and analyse all of them.

A few UNIX files are also useful to investigate a network. These contain the network information required to understand and explore the configuration. */etc/nsswitch.conf* and */etc/resolv.conf* are two useful configuration system text files and can be viewed using the *cat* command call. Both the files reside on the */etc* directory. A listing of different files containing the data to resolve network configuration is available in the *nsswitch.conf* file, whereas DNS name server information is available in the *resolv.conf* file.

```
>cat /etc/nisswitch.conf
...
bootparams: nisplus [NOTFOUND=return] files
```

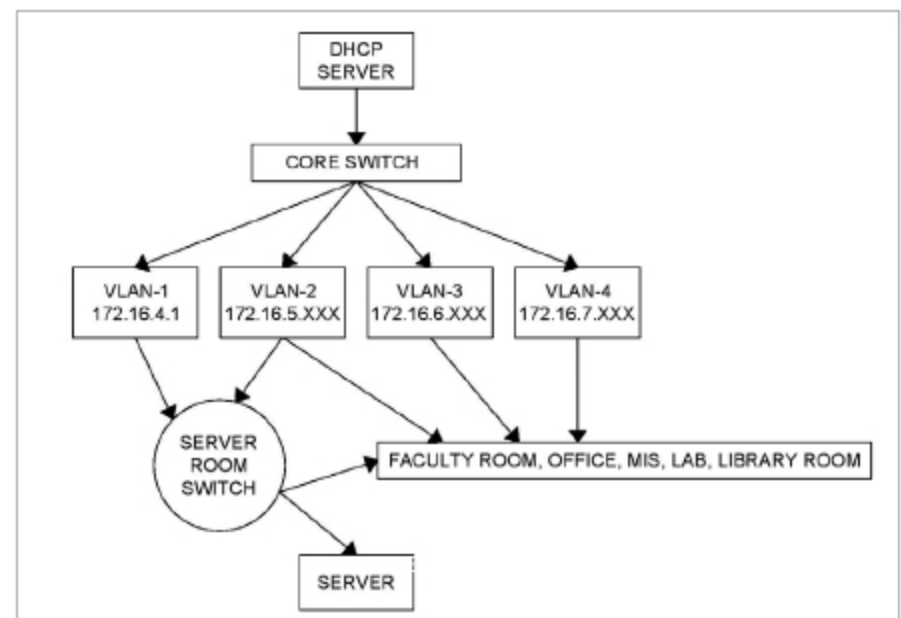


Figure 1: Schematic diagram of a computer network system

```
ethers: files
netmasks: files
networks: files
protocols: files
rpc: files
services: files sss
netgroup: files sss
publickey: nisplus
automount: files
aliases: files nisplus
```

```
>cat /etc/resolv.conf
# Generated by NetworkManager
search ism.int
nameserver 172.16.8.9
nameserver 202.56.230.2
nameserver 8.8.8.8
```

After getting to know the configuration of your own network, it is essential to go beyond and learn about what is within the network, and how to reach there. Networks outside your masked network may be within your own organisation, in different sections or departments; or on outside public networks such as the Internet. To view your network and beyond, it is necessary to see the router table, for which *netstat -r* is a good option.

Here is an example of the *netstat -r* command on a CentOS platform. The default router shows the gateway used to route packets to the outside network. In the absence of the proper functioning of the present name server, the gateway option shows the IP addresses instead of the complete qualified route. In the following example, *netstat -r* is showing the IP router table of the above mentioned network.

```
>netstat -r
Kernel IP routing table
Destination Gateway Genmask Flags MSS Window irtt Iface
```